Systems and Control Theory Take Home Exam 2026

On Toledo you are assigned one of 20 transfer functions which you will investigate and control in question 2 to 4. Prepare the following tasks and answer the accompanying questions **only** for the system assigned to you.

You need to upload **one .zip folder** to Toledo - no other exotic file extensions are allowed. This zip should contain your *Matlab and Simulink files* and *1 pdf which includes all your calculations, drawings and notes.* These are the only code files and documents you can use at the oral defense.
The deadline is **Wednesday 27 May, before 23h59**.

You will defend your work **individually** at the exam. During your exam you will have a 15-minute oral defense. You **do not** have to prepare a report, slides or a presentation - you will just be asked questions about your design, the theory, your code... Make sure your conclusions are ready, and your figures have clear names.

Bring **your own laptop, fully charged,** to be able to show all the **code** you wrote in Matlab and to run it live at the exam. We can ask you to **adapt and recompile your code**. You should be able to explain every line of code. Bring your **own calculations, drawings and notes**, on computer, tablet or paper. You can only use the files you have uploaded on Toledo.

More details about the establishment of the exam task grade can be found on Toledo, including which cases will result in a 0 for the exam task part.

Part 1: Systems Theory

Question 1

When measuring ECG (heart) activity, the sensors are extremely sensitive to muscle tremor (or contractions). Filtering out these tremor frequencies allows doctors (and software) to see the actual biomedical wave without the noise caused by the patients muscle tremor. This can be done with the **filter** as given in Figure 1.

1. Determine the **transfer function** of the filter from Figure 1.
2. Find (and draw on paper) an **electronic circuit** for this filter as a combination of two second order circuits.
3. To validate your circuit, calculate the transfer function for your circuit using **Kirchhoff's laws**. Plot the **pole-zero map** of the filter.
4. Draw the **Bode plot** of your filter. Is your result the same as the result from the first question? Discuss.

Part 2: Control Theory

For questions 2 to 4, use the transfer function assigned to you, see Toledo for more details.

Question 2

1. Discuss the **(in)stability** of the system. Make **more than 1 plot** to support your answer.
2. Build a **P controller** around the transfer function to stabilize the system.
   (a) First discuss which values you consider by using a **Bode plot**.
   (b) Does the **root locus plot** confirm these values?
   (c) Which P-controller makes the system **marginally stable**?
   (d) Pick one value and convince us why you have chosen that **particular** value for K.

Question 3

1. Build a **PID control loop** around the transfer function. Use the result from the previous question to tune it with the Ziegler-Nichols method.

2. Describe a **use case**, i.e. a real-life example of an application. Keep a customer in mind. Set a specific goal for your step response (think about optimizing the rise time, overshoot, settling time, damping, steady-state error and combinations). Describe in detail what your input and output is.

3. Design a **PID controller** with this use case in mind. Start from the controller of question 3.1 and finetune it with the specific goal for your step response in mind. Reflect critically on your outcome.

Question 4

1. Calculate the maximum time delay which can be added in the feedback loop of the PID controller of question 3.3, **based on a Bode plot**.

2. In Simulink, add the **time delay** in the feedback loop of the PID controller of question 3.3. Make sure to use the right time delay block in Simulink (called Transport Delay). Check for which time delays the closed loop becomes unstable. Link with your answer from question 4.1.

3. For the specific time delay that makes your system marginally stable, **tune the PID controller again** such that the output is desirable even with this delay.

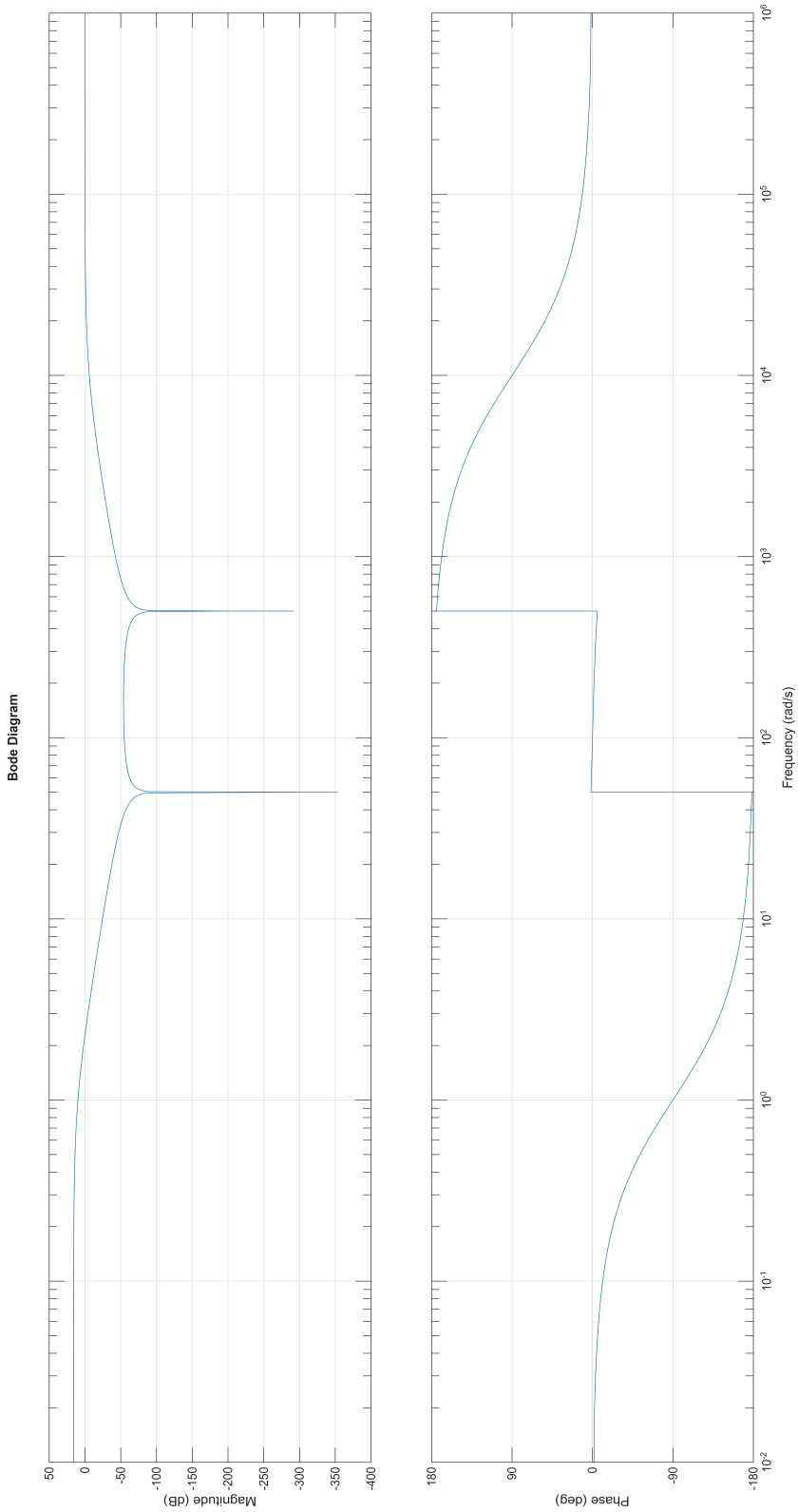

Figure 1: The Bode plot of the unknown filter of Question 1